Mixed Review: Analyzing Functions

TEACHER KEY

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Analyzing a function: combining all 4 skills from this week: family, direction/extrema, domain/range, and symmetry.

Symmetry check: substitute $-x$ into the function and compare the result to $f(x)$ and $-f(x)$.

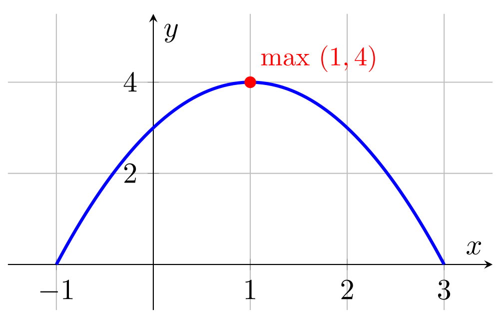
We do together. For $f(x) = -(x-1)^2+4$, find the parent family, max/min with intervals, domain/range, and symmetry.

Step 1. This is a shifted quadratic, so the parent family is quadratic.

Step 2. Vertex form gives vertex $(1, 4)$; the leading coefficient is negative, so it opens down and the vertex is a maximum; increasing $x < 1$, decreasing $x > 1$.

Step 3. There's no root or denominator, so the domain is all reals; the highest output is 4, so the range is $y \leq 4$.

Step 4. Testing symmetry: $f(-x) = -(-x-1)^2 + 4 = -(x+1)^2 + 4$, which matches neither $f(x)$ nor $-f(x)$, so it is neither.



$y = -(x-1)^2 + 4$ — quadratic family, maximum at $(1, 4)$, domain all reals, range $y \leq 4$.

You Try. Solve each. Show every step, then name the answer.

a) For $f(x) = \sqrt{x-3}$, identify the parent family and state the domain and range. square root family; domain $x \geq 3$; range $y \geq 0$

b) Is $f(x) = x^2 + 5$ even, odd, or neither? even: $f(-x) = x^2 + 5 = f(x)$

Summary (fill in): When analyzing any function, work through all 4 checks in order: family, max/min & intervals, domain/range, then symmetry.